

- extra compute and memory (another branch)
- need paired data (image + condition) or condition extraction pipeline

T2I Adapter

- lightweight conditioning adapter for pretrained text-to-image diffusion
 - similar use cases: edges, depth, pose, segmentation, style hints
 - core idea
 - add a small adapter network that encodes the condition into feature maps
 - inject those features into the pretrained U-Net (often via residual connections)
 - typically fewer parameters than ControlNet
 - pretrained diffusion backbone stays frozen
 - how it works (high-level)
 - condition encoder produces multi-scale representations
 - features are fused into U-Net blocks (shallow additions)
 - train only the adapter (and sometimes small projection layers)
 - benefits
 - parameter-efficient, faster to train, lower memory overhead
 - easier to deploy when compute is tight
 - limitations
 - usually weaker controllability than full ControlNet
 - capacity may be insufficient for complex spatial constraints
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Neural Radiance Field (NeRF)

- continuous neural representation of a 3D scene
 - represents a scene as a function parameterized by a neural network

- model the scene as a radiance field
- neural network f_θ
 - input
 - 3D position $x = (x, y, z)$
 - viewing direction d
 - output
 - volume density σ
 - emitted color c
 - training objective
 - minimize photometric reconstruction error
- **Volumetric rendering**
 - pixel color: obtained by integrating color and density along a camera ray
 - encourages consistency across multiple views
- **Positional encoding**
 - $\gamma(x) = (\sin(2^0\pi x), \cos(2^0\pi x), \dots, \sin(2^{L-1}\pi x), \cos(2^{L-1}\pi x))$
 - applies high-frequency Fourier features to inputs
 - maps low-dimensional coordinates into a high-dimensional feature space
 - where linear models (or MLPs in the NTK regime) can express complex functions
 - can be analyzed through the NTK perspective
 - enables the MLP to represent fine details
 - vanilla MLPs exhibit spectral bias → prefer learning low-frequency functions
 - well-adapted to the data which has high-frequency variation
- limitations
 - slow training and rendering
 - requires many views
 - hard to control ($\therefore$ neural network)

Neural Tangent Kernel (NTK)

$$K(x, x') = \nabla_{\theta} f_{\theta}(x)^{\top} \nabla_{\theta} f_{\theta}(x')$$

- a theoretical framework to analyze infinitely wide neural networks
 - as network width $\rightarrow \infty$
 - training dynamics become linear in parameters
 - neural network behaves like a kernel method
 - defines a kernel induced by a neural network architecture
- training interpretation
 - gradient descent on the network = kernel regression with NTK
 - the induced kernel = Neural Tangent Kernel (NTK)
 - explains why overparameterized networks are easy to optimize
- limitations
 - does not fully capture feature learning in finite-width networks
 - more descriptive than predictive for modern deep learning
- **Positional encoding in NeRF**
 - positional encoding fundamentally changes the function space that the MLP can represent
 - applying Fourier features changes the similarity measure between inputs
 - modifies the spectrum of the induced kernel
 - increases sensitivity to high-frequency variations
 - NeRF applies Fourier-based positional encoding to input coordinates
 - maps low-dimensional inputs into a high-dimensional feature space

InstantNGP

- accelerate neural field training and rendering (NeRF-style)
- represent coordinates with a multi-resolution hash grid instead of high-dim sinusoidal encoding

- small MLP predicts density/color from hashed features
 - hash grid gives strong spatial features early → optimization becomes easier
 - multi-resolution captures both coarse structure and fine details
 - limitation
 - hash collisions can introduce artifacts
 - still requires ray marching for rendering
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Gaussian Splatting

- goal: fast, high-quality novel view synthesis without heavy ray marching
- scene representation
 - represent a scene as a set of 3D Gaussians
 - each Gaussian has
 - position, covariance (shape), opacity, color (often with SH coefficients)
- rendering
 - differentiable splatting onto the image plane
 - alpha compositing in depth order
- why good
 - training is efficient and stable
 - rendering is real-time friendly (compared to NeRF)
- limitations
 - needs good initialization (from SfM/colmap-style points)
 - large scenes can require many Gaussians (memory)